Day - 01

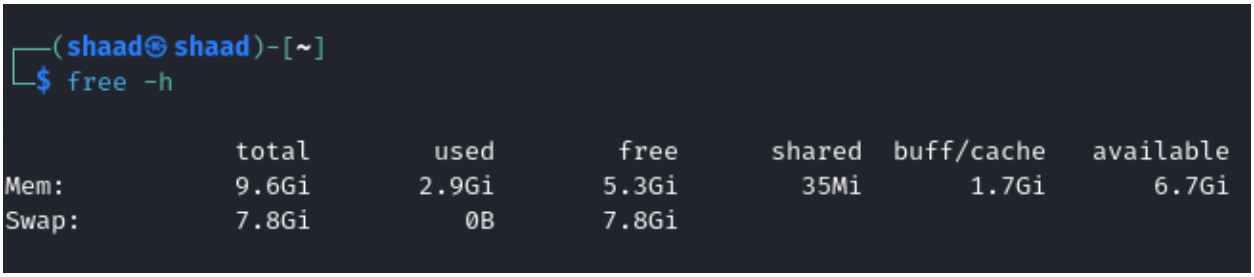
Understanding my System

lscpu : Shows detailed Information about the system's CPU, Architecture,cores, threads, and clock speed.



```
┌──(shaad㉿shaad)-[~]
└─$ lscpu
Architecture:                x86_64
  CPU op-mode(s):            32-bit, 64-bit
  Address sizes:             43 bits physical, 48 bits virtual
  Byte Order:                Little Endian
CPU(s):                      8
  On-line CPU(s) list:       0-7
Vendor ID:                   AuthenticAMD
  Model name:                AMD Ryzen 5 3500U with Radeon Vega Mobile Gfx
    CPU family:              23
    Model:                   24
    Thread(s) per core:      2
    Core(s) per socket:      4
    Socket(s):               1
    Stepping:                1
    Microcode version:       0x8108109
    Frequency boost:         enabled
    CPU(s) scaling MHz:      36%
    CPU max MHz:             3700.0000
    CPU min MHz:             1400.0000
    BogoMIPS:                4192.31
    Flags:                   fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtscp lm constant_tsc rep_good nopl nonstop_tsc cpuid extd_apicid aperf
                             mperf rapl pni pclmulqdq monitor ssse3 fma cx16 sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand lahf_lm cmp_legacy svm extapic cr8_legacy abm sse4a misalignsse 3dnowprefetch osvw skinit wdt tce topoext p
                             erfctr_core perfctr_nb bpext perfctr_llc mwaitx cpb hw_pstate ssbd ibpb vmmcall fsgsbase bmi1 avx2 smep bmi2 rdseed adx smap clflushopt sha_ni xsaveopt xsavec xgetbv1 clzero xsaveerptr arat npt lbrv svm_lock
                             nrip_save tsc_scale vmcb_clean flushbyasid decodeassists pausefilter pfthreshold avic v_vmsave_vmload vgif overflow_recov succor smca sev sev_es
Virtualization features:
  Virtualization:            AMD-V
Caches (sum of all):
  L1d:                       128 KiB (4 instances)
  L1i:                       256 KiB (4 instances)
  L2:                        2 MiB (4 instances)
  L3:                        4 MiB (1 instance)
NUMA:
  NUMA node(s):              1
  NUMA node0 CPU(s):         0-7
Vulnerabilities:
  Gather data sampling:      Not affected
  Ghostwrite:                Not affected
  Indirect target selection: Not affected
  Itlb multihit:             Not affected
  L1tf:                      Not affected
  Mds:                       Not affected
  Meltdown:                  Not affected
  Mmio stale data:           Not affected
  Old microcode:             Not affected
  Reg file data sampling:    Not affected
  Retbleed:                  Mitigation; untrained return thunk; SMT vulnerable
  Spec rstack overflow:      Mitigation; Safe RET
  Spec store bypass:         Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:                Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:                Mitigation; Retpolines; IBPB conditional; STIBP disabled; RSB filling; PBRSB-eIBRS Not affected; BHI Not affected
  Srbds:                     Not affected
  Tsa:                       Not affected
  Tsx async abort:           Not affected
  Vmscape:                   Mitigation; IBPB before exit to userspace
```

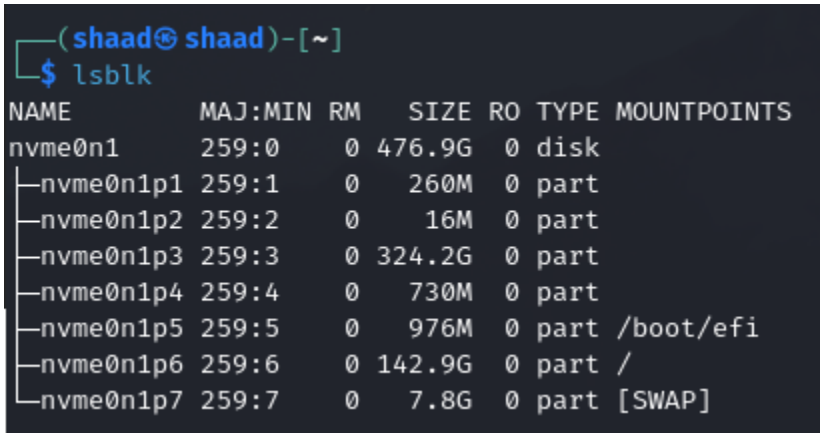
free -h : Shows system's RAM and swap memory usage



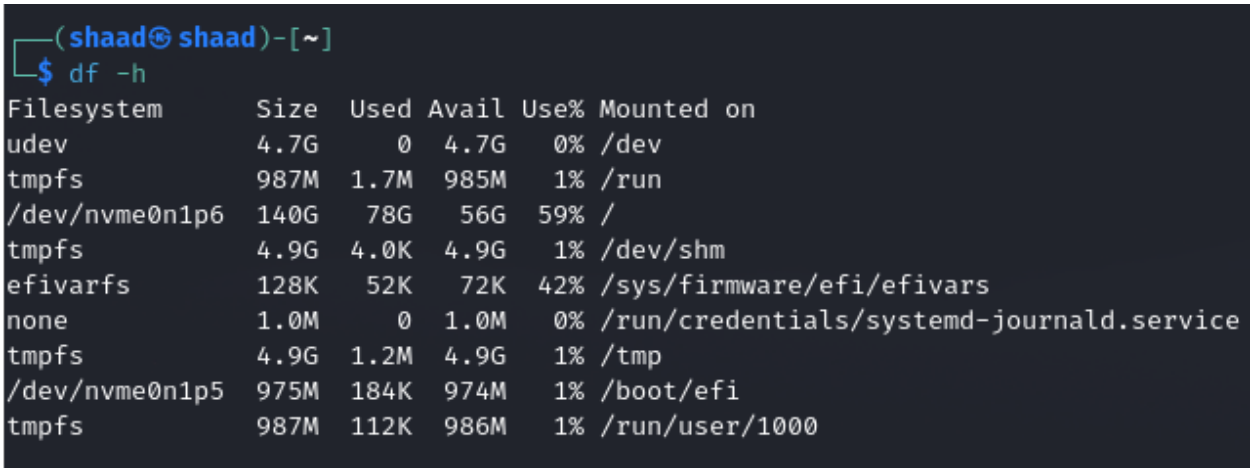
```
┌──(shaad㉿shaad)-[~]
└─$ free -h

               total        used        free      shared  buff/cache   available
Mem:           9.6Gi       2.9Gi       5.3Gi        35Mi       1.7Gi       6.7Gi
Swap:          7.8Gi          0B       7.8Gi
```

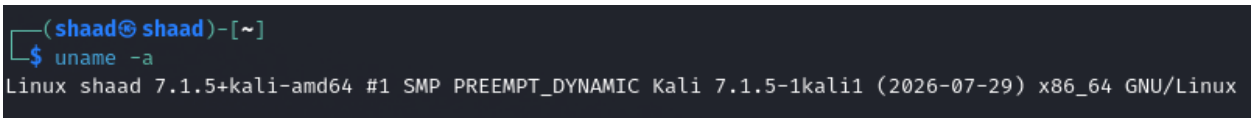
lsblk : lists all storage devices such as hard drives , ssds and their partitions



Df -h : shows disk space usage of mounted file system



Uname -a : shows system information



Exploring Hardware

Lsusb : shows all USB devices connected to the system

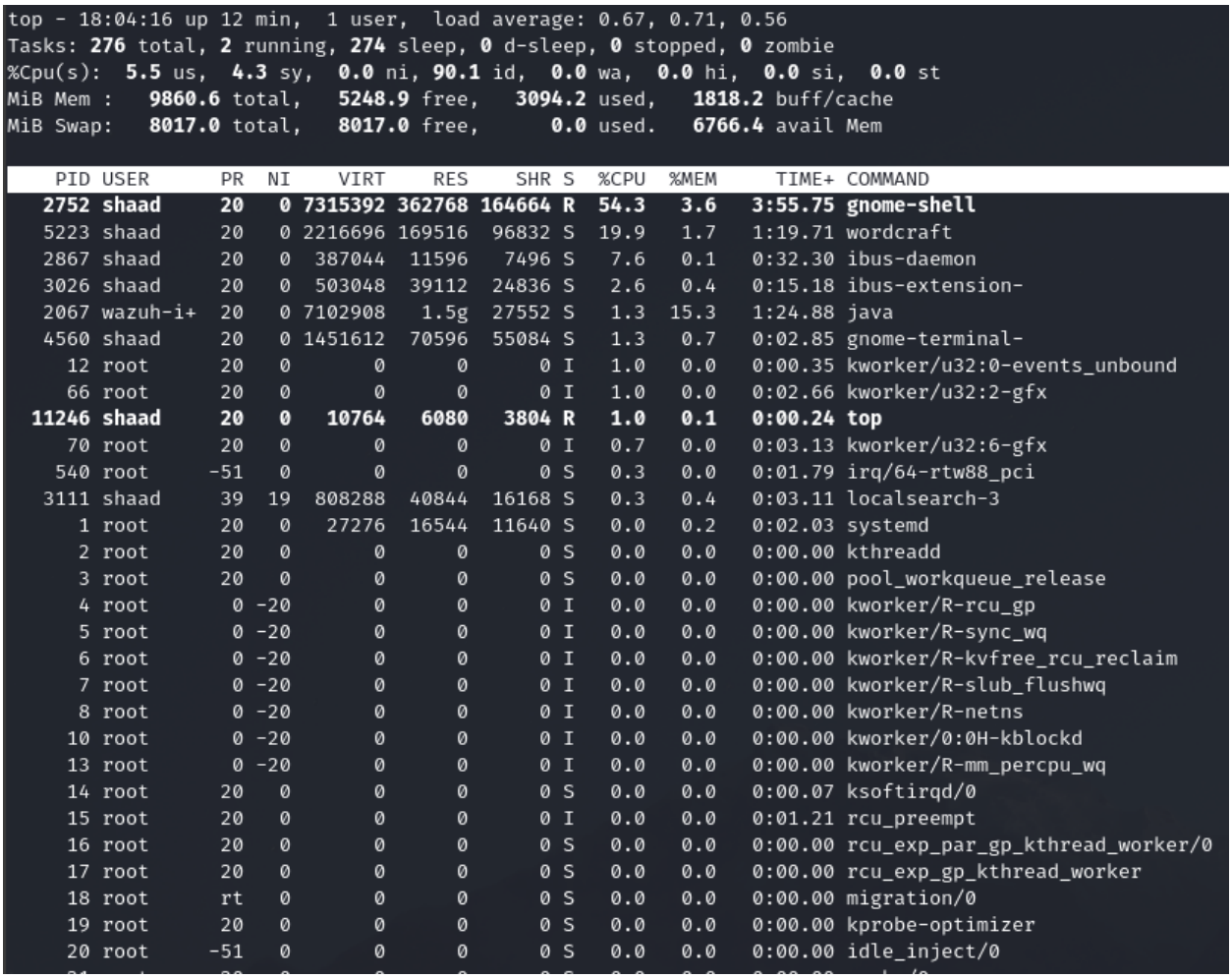
```
┌──(shaad㉿shaad)-[~]
└─$ lsusb
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 001 Device 002: ID 046d:c542 Logitech, Inc. M185 compact wireless mouse
Bus 001 Device 003: ID 0bda:b00a Realtek Semiconductor Corp. Realtek Bluetooth 4.2 Adapter
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 003 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 003 Device 002: ID 05e3:0608 Genesys Logic, Inc. Hub
Bus 003 Device 003: ID 30c9:0013 Luxvisions Innotech Limited HP TrueVision HD Camera
Bus 004 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
```

Lspci : shows all PCI devices connected to the system

```
┌──(shaad㉿shaad)-[~]
└─$ lspci
00:00.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Raven/Raven2 Root Complex
00:00.2 IOMMU: Advanced Micro Devices, Inc. [AMD] Raven/Raven2 IOMMU
00:01.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Family 17h (Models 00h-1fh) PCIe Dummy Host Bridge
00:01.2 PCI bridge: Advanced Micro Devices, Inc. [AMD] Raven/Raven2 PCIe GPP Bridge [6:0]
00:01.6 PCI bridge: Advanced Micro Devices, Inc. [AMD] Raven/Raven2 PCIe GPP Bridge [6:0]
00:01.7 PCI bridge: Advanced Micro Devices, Inc. [AMD] Raven/Raven2 PCIe GPP Bridge [6:0]
00:08.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Family 17h (Models 00h-1fh) PCIe Dummy Host Bridge
00:08.1 PCI bridge: Advanced Micro Devices, Inc. [AMD] Raven/Raven2 Internal PCIe GPP Bridge 0 to Bus A
00:14.0 SMBus: Advanced Micro Devices, Inc. [AMD] FCH SMBus Controller (rev 61)
00:14.3 ISA bridge: Advanced Micro Devices, Inc. [AMD] FCH LPC Bridge (rev 51)
00:18.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Raven/Raven2 Device 24: Function 0
00:18.1 Host bridge: Advanced Micro Devices, Inc. [AMD] Raven/Raven2 Device 24: Function 1
00:18.2 Host bridge: Advanced Micro Devices, Inc. [AMD] Raven/Raven2 Device 24: Function 2
00:18.3 Host bridge: Advanced Micro Devices, Inc. [AMD] Raven/Raven2 Device 24: Function 3
00:18.4 Host bridge: Advanced Micro Devices, Inc. [AMD] Raven/Raven2 Device 24: Function 4
00:18.5 Host bridge: Advanced Micro Devices, Inc. [AMD] Raven/Raven2 Device 24: Function 5
00:18.6 Host bridge: Advanced Micro Devices, Inc. [AMD] Raven/Raven2 Device 24: Function 6
00:18.7 Host bridge: Advanced Micro Devices, Inc. [AMD] Raven/Raven2 Device 24: Function 7
01:00.0 Non-Volatile memory controller: Samsung Electronics Co Ltd NVMe SSD Controller SM981/PM981/PM983
02:00.0 Network controller: Realtek Semiconductor Co., Ltd. RTL8821CE 802.11ac PCIe Wireless Network Adapter
04:00.0 VGA compatible controller: Advanced Micro Devices, Inc. [AMD/ATI] Picasso/Raven 2 [Radeon Vega Series / Radeon Vega Mobile Series] (rev c2)
04:00.1 Audio device: Advanced Micro Devices, Inc. [AMD/ATI] Raven/Raven2/Fenghuang HDMI/DP Audio Controller
04:00.2 Encryption controller: Advanced Micro Devices, Inc. [AMD] Raven/Raven2/FireFlight/Renoir/Cezanne Platform Security Processor
04:00.3 USB controller: Advanced Micro Devices, Inc. [AMD] Raven USB 3.1
04:00.4 USB controller: Advanced Micro Devices, Inc. [AMD] Raven USB 3.1
04:00.5 Multimedia controller: Advanced Micro Devices, Inc. [AMD] Audio Coprocessor
04:00.6 Audio device: Advanced Micro Devices, Inc. [AMD] Ryzen HD Audio Controller
```

Process Basics

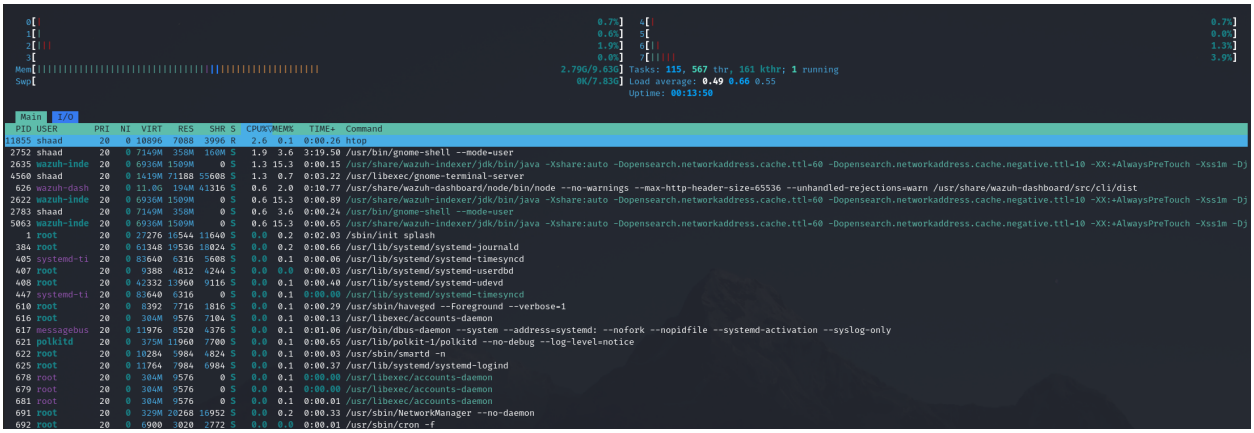
Top : shows running processes , CPU usage, memory usage and system load



```
top - 18:04:16 up 12 min,  1 user,  load average: 0.67, 0.71, 0.56
Tasks: 276 total, 2 running, 274 sleep, 0 d-sleep, 0 stopped, 0 zombie
%Cpu(s):  5.5 us,  4.3 sy,  0.0 ni, 90.1 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   9860.6 total,   5248.9 free,   3094.2 used,   1818.2 buff/cache
MiB Swap:   8017.0 total,   8017.0 free,      0.0 used.   6766.4 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   2752 shaad     20   0 7315392 362768 164664 R  54.3   3.6   3:55.75 gnome-shell
   5223 shaad     20   0 2216696 169516  96832 S  19.9   1.7   1:19.71 wordcraft
   2867 shaad     20   0  387044  11596   7496 S   7.6   0.1   0:32.30 ibus-daemon
   3026 shaad     20   0  503048  39112  24836 S   2.6   0.4   0:15.18 ibus-extension-
   2067 wazuh-i+  20   0 7102908   1.5g  27552 S   1.3  15.3   1:24.88 java
   4560 shaad     20   0 1451612  70596  55084 S   1.3   0.7   0:02.85 gnome-terminal-
     12 root      20   0       0      0      0 I   1.0   0.0   0:00.35 kworker/u32:0-events_unbound
     66 root      20   0       0      0      0 I   1.0   0.0   0:02.66 kworker/u32:2-gfx
  11246 shaad     20   0   10764   6080   3804 R   1.0   0.1   0:00.24 top
     70 root      20   0       0      0      0 I   0.7   0.0   0:03.13 kworker/u32:6-gfx
    540 root     -51   0       0      0      0 S   0.3   0.0   0:01.79 irq/64-rtw88_pci
   3111 shaad     39  19  808288  40844  16168 S   0.3   0.4   0:03.11 localsearch-3
      1 root      20   0   27276  16544  11640 S   0.0   0.2   0:02.03 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-kblockd
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 S   0.0   0.0   0:00.07 ksoftirqd/0
     15 root      20   0       0      0      0 I   0.0   0.0   0:01.21 rcu_preempt
     16 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
     17 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_gp_kthread_worker
     18 root      rt   0       0      0      0 S   0.0   0.0   0:00.00 migration/0
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kprobe-optimizer
     20 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
```

Htop : shows system performance and running process in an interactive user friendly interface

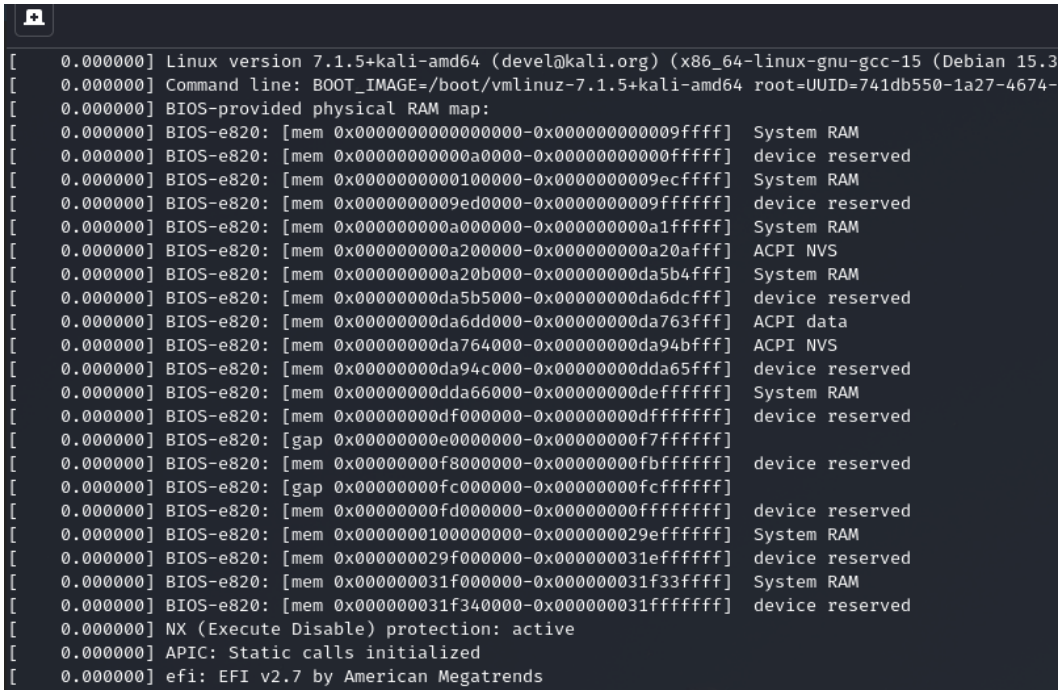


Ps aux : shows currently running processes including users, CPU and memory usage

```
┌──(shaad㉿shaad)-[~]
└─$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.2  0.1  27276 16544 ?        Ss   17:51   0:02 /sbin/init splash
root           2  0.0  0.0      0     0 ?        S    17:51   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    17:51   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   17:51   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   17:51   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   17:51   0:00 [kworker/R-kvfree_rcu_reclaim]
root           7  0.0  0.0      0     0 ?        I<   17:51   0:00 [kworker/R-slub_flushwq]
root           8  0.0  0.0      0     0 ?        I<   17:51   0:00 [kworker/R-netns]
root          10  0.0  0.0      0     0 ?        I<   17:51   0:00 [kworker/0:0H-kblockd]
root          12  0.0  0.0      0     0 ?        I    17:51   0:00 [kworker/u32:0-phy0]
root          13  0.0  0.0      0     0 ?        I<   17:51   0:00 [kworker/R-mm_percpu_wq]
root          14  0.0  0.0      0     0 ?        S    17:51   0:00 [ksoftirqd/0]
root          15  0.1  0.0      0     0 ?        I    17:51   0:01 [rcu_preempt]
root          16  0.0  0.0      0     0 ?        S    17:51   0:00 [rcu_exp_par_gp_kthread_worker/0]
root          17  0.0  0.0      0     0 ?        S    17:51   0:00 [rcu_exp_gp_kthread_worker]
root          18  0.0  0.0      0     0 ?        S    17:51   0:00 [migration/0]
root          19  0.0  0.0      0     0 ?        S    17:51   0:00 [kprobe-optimizer]
root          20  0.0  0.0      0     0 ?        S    17:51   0:00 [idle_inject/0]
root          21  0.0  0.0      0     0 ?        S    17:51   0:00 [cpuhp/0]
root          22  0.0  0.0      0     0 ?        S    17:51   0:00 [cpuhp/2]
root          23  0.0  0.0      0     0 ?        S    17:51   0:00 [idle_inject/2]
root          24  0.0  0.0      0     0 ?        S    17:51   0:00 [migration/2]
root          25  0.0  0.0      0     0 ?        S    17:51   0:00 [ksoftirqd/2]
root          27  0.0  0.0      0     0 ?        I<   17:51   0:00 [kworker/2:0H-kblockd]
root          28  0.0  0.0      0     0 ?        S    17:51   0:00 [cpuhp/4]
```

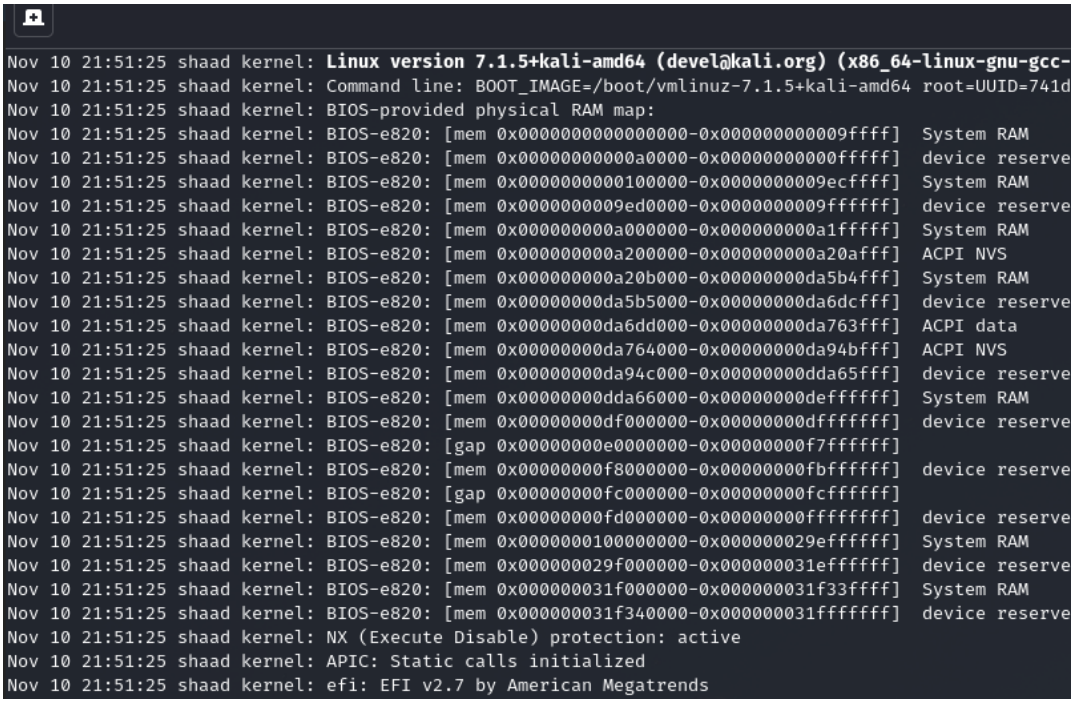
Boot & Logs

Dmesg | less: shows kernel messages



```
[    0.000000] Linux version 7.1.5+kali-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-15 (Debian 15.3
[    0.000000] Command line: BOOT_IMAGE=/boot/vmlinuz-7.1.5+kali-amd64 root=UUID=741db550-1a27-4674-
[    0.000000] BIOS-provided physical RAM map:
[    0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff]  System RAM
[    0.000000] BIOS-e820: [mem 0x00000000000a0000-0x00000000000fffff]  device reserved
[    0.000000] BIOS-e820: [mem 0x0000000000100000-0x0000000009ecffff]  System RAM
[    0.000000] BIOS-e820: [mem 0x0000000009ed0000-0x0000000009ffffff]  device reserved
[    0.000000] BIOS-e820: [mem 0x000000000a000000-0x000000000a1fffff]  System RAM
[    0.000000] BIOS-e820: [mem 0x000000000a200000-0x000000000a20afff]  ACPI NVS
[    0.000000] BIOS-e820: [mem 0x000000000a20b000-0x00000000da5b4fff]  System RAM
[    0.000000] BIOS-e820: [mem 0x00000000da5b5000-0x00000000da6dcfff]  device reserved
[    0.000000] BIOS-e820: [mem 0x00000000da6dd000-0x00000000da763fff]  ACPI data
[    0.000000] BIOS-e820: [mem 0x00000000da764000-0x00000000da94bfff]  ACPI NVS
[    0.000000] BIOS-e820: [mem 0x00000000da94c000-0x00000000dda65fff]  device reserved
[    0.000000] BIOS-e820: [mem 0x00000000dda66000-0x00000000deffffff]  System RAM
[    0.000000] BIOS-e820: [mem 0x00000000df000000-0x00000000dfffffff]  device reserved
[    0.000000] BIOS-e820: [gap 0x00000000e0000000-0x00000000f7ffffff]
[    0.000000] BIOS-e820: [mem 0x00000000f8000000-0x00000000fbffffff]  device reserved
[    0.000000] BIOS-e820: [gap 0x00000000fc000000-0x00000000fcffffff]
[    0.000000] BIOS-e820: [mem 0x00000000fd000000-0x00000000ffffffff]  device reserved
[    0.000000] BIOS-e820: [mem 0x0000000100000000-0x000000029effffff]  System RAM
[    0.000000] BIOS-e820: [mem 0x000000029f000000-0x000000031effffff]  device reserved
[    0.000000] BIOS-e820: [mem 0x000000031f000000-0x000000031f33ffff]  System RAM
[    0.000000] BIOS-e820: [mem 0x000000031f340000-0x000000031fffffff]  device reserved
[    0.000000] NX (Execute Disable) protection: active
[    0.000000] APIC: Static calls initialized
[    0.000000] efi: EFI v2.7 by American Megatrends
```

Journalctl -xb : shows system logs from the current boot session , including errors and troubleshooting information



```
Nov 10 21:51:25 shaad kernel: Linux version 7.1.5+kali-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-
Nov 10 21:51:25 shaad kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-7.1.5+kali-amd64 root=UUID=741d
Nov 10 21:51:25 shaad kernel: BIOS-provided physical RAM map:
Nov 10 21:51:25 shaad kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff]  System RAM
Nov 10 21:51:25 shaad kernel: BIOS-e820: [mem 0x00000000000a0000-0x00000000000fffff]  device reserve
Nov 10 21:51:25 shaad kernel: BIOS-e820: [mem 0x0000000000100000-0x0000000009ecffff]  System RAM
Nov 10 21:51:25 shaad kernel: BIOS-e820: [mem 0x0000000009ed0000-0x0000000009ffffff]  device reserve
Nov 10 21:51:25 shaad kernel: BIOS-e820: [mem 0x000000000a000000-0x000000000a1fffff]  System RAM
Nov 10 21:51:25 shaad kernel: BIOS-e820: [mem 0x000000000a200000-0x000000000a20afff]  ACPI NVS
Nov 10 21:51:25 shaad kernel: BIOS-e820: [mem 0x000000000a20b000-0x00000000da5b4fff]  System RAM
Nov 10 21:51:25 shaad kernel: BIOS-e820: [mem 0x00000000da5b5000-0x00000000da6dcfff]  device reserve
Nov 10 21:51:25 shaad kernel: BIOS-e820: [mem 0x00000000da6dd000-0x00000000da763fff]  ACPI data
Nov 10 21:51:25 shaad kernel: BIOS-e820: [mem 0x00000000da764000-0x00000000da94bfff]  ACPI NVS
Nov 10 21:51:25 shaad kernel: BIOS-e820: [mem 0x00000000da94c000-0x00000000dda65fff]  device reserve
Nov 10 21:51:25 shaad kernel: BIOS-e820: [mem 0x00000000dda66000-0x00000000deffffff]  System RAM
Nov 10 21:51:25 shaad kernel: BIOS-e820: [mem 0x00000000df000000-0x00000000dfffffff]  device reserve
Nov 10 21:51:25 shaad kernel: BIOS-e820: [gap 0x00000000e0000000-0x00000000f7ffffff]
Nov 10 21:51:25 shaad kernel: BIOS-e820: [mem 0x00000000f8000000-0x00000000fbffffff]  device reserve
Nov 10 21:51:25 shaad kernel: BIOS-e820: [gap 0x00000000fc000000-0x00000000fcffffff]
Nov 10 21:51:25 shaad kernel: BIOS-e820: [mem 0x00000000fd000000-0x00000000ffffffff]  device reserve
Nov 10 21:51:25 shaad kernel: BIOS-e820: [mem 0x0000000100000000-0x000000029effffff]  System RAM
Nov 10 21:51:25 shaad kernel: BIOS-e820: [mem 0x000000029f000000-0x000000031effffff]  device reserve
Nov 10 21:51:25 shaad kernel: BIOS-e820: [mem 0x000000031f000000-0x000000031f33ffff]  System RAM
Nov 10 21:51:25 shaad kernel: BIOS-e820: [mem 0x000000031f340000-0x000000031fffffff]  device reserve
Nov 10 21:51:25 shaad kernel: NX (Execute Disable) protection: active
Nov 10 21:51:25 shaad kernel: APIC: Static calls initialized
Nov 10 21:51:25 shaad kernel: efi: EFI v2.7 by American Megatrends
```

Hands on Task

CPU model : AMD Ryzen 5 3500U with Radeon Vega Mobile Gfx

```
┌──(shaad㉿shaad)-[~]
└─$ lscpu
Architecture:                         x86_64
  CPU op-mode(s):                     32-bit, 64-bit
  Address sizes:                      43 bits physical, 48 bits virtual
  Byte Order:                         Little Endian
CPU(s):                               8
  On-line CPU(s) list:                0-7
Vendor ID:                            AuthenticAMD
  Model name:                         AMD Ryzen 5 3500U with Radeon Vega Mobile Gfx
    CPU family:                       23
    Model:                            24
    Thread(s) per core:               2
    Core(s) per socket:               4
    Socket(s):                        1
    Stepping:                         1
```

RAM Size : 9.6 GB

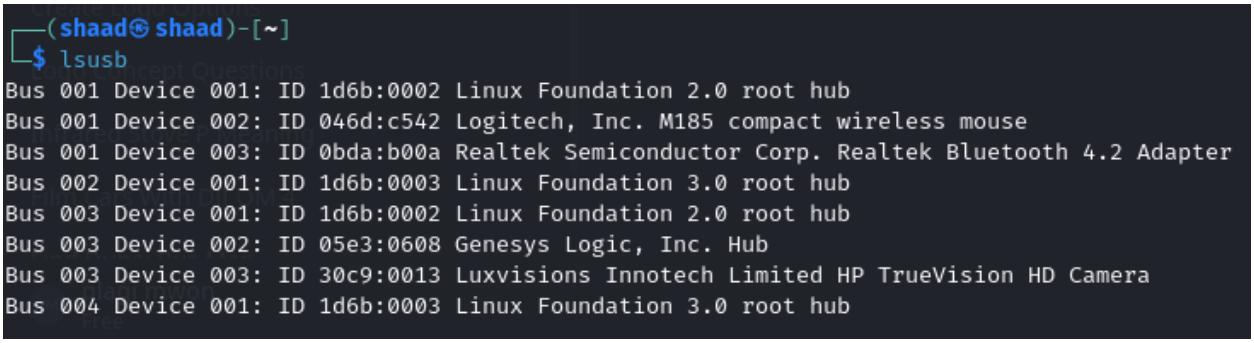
```
┌──(shaad㉿shaad)-[~]
└─$ free -h

               total        used        free      shared  buff/cache   available
Mem:           9.6Gi       2.9Gi       5.3Gi        35Mi       1.7Gi       6.7Gi
Swap:          7.8Gi          0B       7.8Gi
```

Disk Partition : 476.9 GB divided into 7 partition

```
┌──(shaad㉿shaad)-[~]
└─$ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
nvme0n1     259:0    0 476.9G  0 disk
├─nvme0n1p1 259:1    0   260M  0 part
├─nvme0n1p2 259:2    0    16M  0 part
├─nvme0n1p3 259:3    0 324.2G  0 part
├─nvme0n1p4 259:4    0   730M  0 part
├─nvme0n1p5 259:5    0   976M  0 part /boot/efi
├─nvme0n1p6 259:6    0 142.9G  0 part /
└─nvme0n1p7 259:7    0   7.8G  0 part [SWAP]
```

USB devices : 8 USB entries



How a System boots?

1.Power on : CPU receives power and starts executing firmware instructions

2.BIOS/UEFI : Initializes hardware components and performs POST (Power On Self Test)

3.Bootloader: The firmware identifies a bootable device and loads the bootloader

4.Kernel Loading: The bootloader loads the operating system kernel into RAM.

5.Init/Systemd: The kernel initializes hardware, drivers and starts the systemd initialization process.

6.Login: System services start, and the login screen or command-line interface appears.